

Mechanical Ventilation during ECMO: Best Practices

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Adults and children who require extracorporeal membrane oxygenation for respiratory failure remain at risk for ongoing lung injury if ventilator management is not optimized. This review serves as a guide to assist the bedside clinician in ventilator titration for patients on extracorporeal membrane oxygenation, with a focus on lung-protective strategies. Existing data and guidelines for extracorporeal membrane oxygenation ventilator management are reviewed, including non-conventional ventilation modes and adjunct therapies. Key words: Mechanical ventilation; extracorporeal membrane oxygenation; ventilator induced lung injury; acute respiratory distress syndrome. [Respir Care 2023;68(6):838–845. © 2023 Daedalus Enterprises]

Introduction

The use of extracorporeal membrane oxygenation (ECMO) as a therapy for refractory hypoxemic respiratory failure continues to increase, and ongoing technological and strategic improvements have been associated with improved ECMO survival.^{1–4} Yet, despite any successes with ECMO, it must be acknowledged that ECMO is not a cure, and, in lieu of bridging a patient to transplantation, ECMO only buys time to allow a patient to recover from his or her underlying disease process. As such, it is imperative for the ECMO practitioner to focus on strategies that support

recovery and minimize potential harms from necessary therapies. For the patient on respiratory ECMO, avoiding ventilator-induced lung injury (VILI) is paramount to optimize chances for lung recovery.

Although there are many potential ventilator management approaches for patients supported on ECMO, the overarching principle of lung protection is essential to give the patient the best chance to recover with minimal morbidity. The same lung-protective strategies that have been demonstrated to reduce mortality from severe ARDS still hold true once the patient is placed on ECMO, and the addition of the oxygenator, which acts as a third lung, allows the clinician to reduce ventilator support to meet lung-protective targets. This review seeks to provide a framework to help the bedside clinician titrate the ventilator for patients on ECMO for respiratory failure to meet clinical targets and still support recovery. We review the basic tenets of lung protection, available data and guidelines for ventilator management of patients on ECMO, and briefly investigate alternative modes of ventilation and ancillary therapies.

Principles of Lung Protection

To be able to optimize lung-protective strategies on ECMO, it is important to understand the multiple pathways

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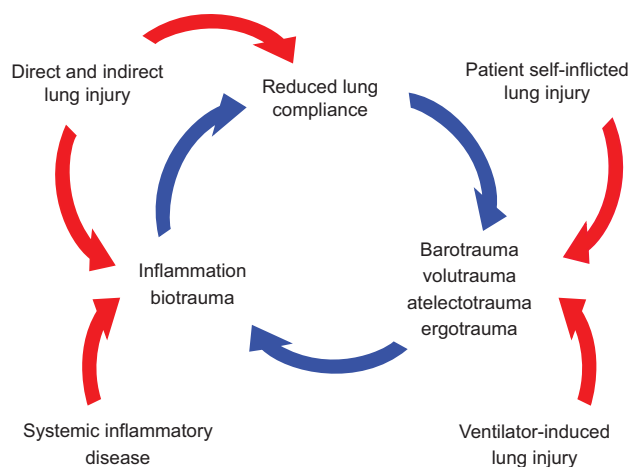


Fig. 1. Pathophysiology of lung injury.

of VILI common in ARDS: barotrauma, volutrauma, atelectotrauma, ergotrauma, and biotrauma. These different types of injury are not completely independent of one another; their relationship is modeled in Figure 1. Volutrauma and barotrauma are directly related through the patient's lung compliance because increased driving pressures correlate with larger tidal volumes (V_T), each of which causes direct stress on type 1 alveolar cells and endothelial basement membrane that line the alveoli. It has been long understood that targeting low V_T (~ 6 mL/kg ideal body weight) and limiting plateau pressure to ≤ 30 cm H₂O improves mortality in severe ARDS, through limiting both tidal and maximum alveolar stretch.⁵⁻⁷ More recent data further support that limiting driving pressure has a beneficial effect on mortality independent of the level of PEEP or peak inspiratory pressure,⁸ which further supports the concept of limiting tidal stretch and total mechanical energy delivered to the lung, termed ergotrauma.⁹

Due to heterogeneous lung disease, atelectotrauma is also common in ARDS and arises from a cycle of repeated collapse and reopening of alveoli. Shear injury during this cycle triggers cellular damage, death, and inflammation. Atelectotrauma can be mitigated through an open lung strategy, most commonly achieved through adequate PEEP such that there is minimum alveolar collapse between breaths. Open lung strategy has additional benefits by maintaining more homogenous lung compliance throughout a delivered breath, with more equal delivery of V_T across a larger portion of the lung and less shear from lung distortion between open and collapsed lung units.

Avoiding biotrauma is largely dependent on treating any underlying disease process that causes a hyperinflammatory state but is also dependent on reducing ongoing inflammation secondary to trauma to the lungs, including avoiding VILI. Exposure to high levels of F_{IO_2} is another source of inflammation due to oxidative stress and free radical formation.^{10,11}

The addition of an ECMO circuit in itself is also known to increase inflammation through vessel injury, exposure of blood to a foreign body in the form of the artificial components of the circuit, and through shear injury to cells by the pump and areas of turbulent flow.¹² Despite this, Rozenchwajg et al¹³ reported a reduction in multiple serum inflammatory markers after subjects who were critically ill with ARDS were transitioned to ECMO support with lung-protective ventilation, and a porcine model of ARDS treated with ECMO showed less histologic evidence of lung injury with increasing lung-protective strategies.¹⁴ Each of these studies demonstrate the importance of lung protection on reducing biotrauma in patients on ECMO.

It is also important to recognize that there is no need to target completely normal gas exchange for a patient on ECMO for refractory respiratory failure. Permissive hypercapnia and permissive hypoxia are proven strategies to reduce ventilator settings and ergotrauma¹⁵⁻¹⁷ as well as reducing F_{IO_2} to minimize oxidative stress. In the setting of adequate oxygen delivery to tissues, patients will often tolerate saturations in the 80s without deleterious effects. Similarly, in lieu of intracranial hypertension or significant pulmonary hypertension, patients will tolerate markedly elevated P_{CO_2} as long as pH is maintained at ≥ 7.2 .

Ventilator Targets

Without clear data driving specific targets for ventilator settings on ECMO, it can be useful to look at established guidelines and protocols used in clinical trials. Guidelines for mechanical ventilation during ECMO from the Extracorporeal Life Support Organization as well as the ECMO conventional ventilator protocol settings for the recent EOLIA (ECMO to Rescue Lung Injury in Severe ARDS) and CESAR (Conventional Ventilation or ECMO For Severe Adult Respiratory Failure) randomized control trials of ECMO versus conventional therapy for refractory ARDS are presented in Table 1.^{1,18,19}

These recommended ventilator settings are notable in that they comply with generally accepted lung-protective ventilator settings for ARDS; specifically plateau pressure ≤ 30 cm H₂O, $F_{IO_2} \leq 0.60$, and PEEP ≥ 10 cm H₂O. Although V_T is not specified in these guidelines, the recommended pressure settings coupled with the poor compliance of patients on ECMO for ARDS would generally limit V_T to the ≤ 6 mL/kg ideal body weight that would be considered lung protective. However, if the patient's lung compliance is higher than that of the typical patient with ARDS and on ECMO, the ventilator should be weaned to target V_T of approximately 6 mL/kg ideal body weight, and to minimize driving pressure. Similar to patients with ARDS who are not on ECMO, there are data that show reduced biotrauma and improved survival for patients on ECMO when lower driving pressure is used.^{20,21} In a meta-analysis of

Table 1. ELSO and EOLIA/CESAR Guidelines for Conventional Ventilator Settings

Setting	ELSO Acceptable Range	ELSO Recommended Range	EOLIA Protocol	CESAR Trial
Inspiratory plateau pressure, cm H ₂ O	≤30	<25	≤24	20
Tidal volume	NR	NR	Adjusted to meet plateau pressure target	NR
PEEP, cm H ₂ O	≥10	10–24	≥10	10
F _{IO₂}	0.30–0.50	As low as possible to maintain saturations	0.30–0.50	0.30–0.50
Breathing frequency, breaths/min	4–30	4–15	4–30	10

ELSO = Extracorporeal Life Support Organization¹⁹
 EOLIA = ECMO to Rescue Lung Injury in Severe ARDS¹⁸
 CESAR = Conventional Ventilation or ECMO For Severe Adult Respiratory Failure¹
 NR = not reported

9 studies and 545 subjects on ECMO, a lower driving pressure was significantly associated with survival.²¹

The effectiveness of carbon dioxide removal by the extracorporeal membrane oxygenator allows the practitioner to significantly wean the patient's minute ventilation, which allows for small or even negligible V_T in lungs with low compliance. Breathing frequency may also be reduced to further reduce the mechanical work applied to the lungs; frequency rates of ~ 8–10 breaths/min are common in all ages while on ECMO. It is important that practitioners deter from making changes on the ventilator to improve gas exchange but instead adjust sweep gas flow to improve ventilation and ECMO blood flow to improve oxygenation.

For patients on ECMO, It is not clear if ultra-lung-protective strategies of $V_T < 4$ mL/kg ideal body weight, driving pressure < 15 cm H₂O, and frequency < 10 breaths/min provide additional benefit beyond the traditional lung-protective strategies mentioned above, yet this is a common strategy used in patients. Schmidt et al²² reported that 39% of adult patients on ECMO were ventilated with ultra-protective settings. One randomized trial of patients with ARDS and on venovenous ECMO found no advantage to ultra-lung-protective strategies of V_T 1–2 mL/kg ideal body weight and frequency of 5–10 breaths/min when compared with the EOLIA protocol detailed in Table 1.²³

PEEP should be titrated to optimize lung expansion without negatively affecting hemodynamics. Although a majority of patients on ECMO may have significant atelectasis early in their ECMO course, an open lung strategy will not only assist with weaning the patient toward liberation from the ECMO circuit²⁴ but will also help avoid VILI. Recruitment of closed alveoli will improve compliance, allow any delivered V_T to be distributed more evenly across lung units, and avoid the alveolar collapse-reopen cycle that contributes to atelectotrauma. Although the guidelines mentioned above suggest a PEEP of at least 10 cm H₂O, PEEPs of 12–16 cm H₂O are common in patients on ECMO, and higher PEEPs may be indicated in many patients. Assessing lung expansion through chest radiographs, ventilator graphics such as

pressure volume loops or stress index, esophageal manometry, or electrical impedance may all assist the bedside practitioner in titrating PEEP to an optimal level. The ability to tolerate low driving pressures and small V_T will allow high levels of PEEP without requiring high plateau pressures.

An alternative PEEP strategy involves weaning PEEP toward extubation while continuing ECMO to support spontaneous breathing, weaning of sedation, and pulmonary rehabilitation.^{25,26} Yet, these subjects were typically weaned and extubated several days into their ECMO course, after lungs had been recruited and were at least partially open. Other case reports exist that demonstrate successful use of ECMO to avoid intubation completely in patients with ARDS,^{27,28} but these patients are often selected carefully for their eligibility for this strategy, and no data exist that demonstrate comparative outcomes for this strategy.

As the patient improves, weaning sedation to allow the patient to take spontaneous breaths has several potential advantages, including improved ventilation/perfusion matching, lung recruitment, secretion mobilization, and decreased diaphragmatic myopathy. However, with an increase in spontaneous breathing, attention should be paid to optimizing patient-ventilator synchrony, avoiding asynchrony that may contribute to VILI, including double triggering, reverse triggering, and cycle delay. Patient self-inflicted lung injury may also result from the patient's natural respiratory drive and leads him or her to take abnormally large breaths, typically in the setting of air hunger. Similar injury that occurs due to ventilator-induced volutrauma and atelectotrauma can occur with these large patient-initiated spontaneous breaths, with strain on the alveolar cells and shear injury at the interface between open and collapsed alveoli and lung units.^{29–31} Premature weaning of sweep gas may contribute to this air hunger and self-inflicted injury, whereas increasing sweep gas may alleviate this condition in many patients without having to increase sedation.^{32,33} There generally should be no urgency to aggressively wean sweep gas until the patient has mostly recovered and is close to decannulation; attention should rather be paid to weaning ventilator driving pressure as able.

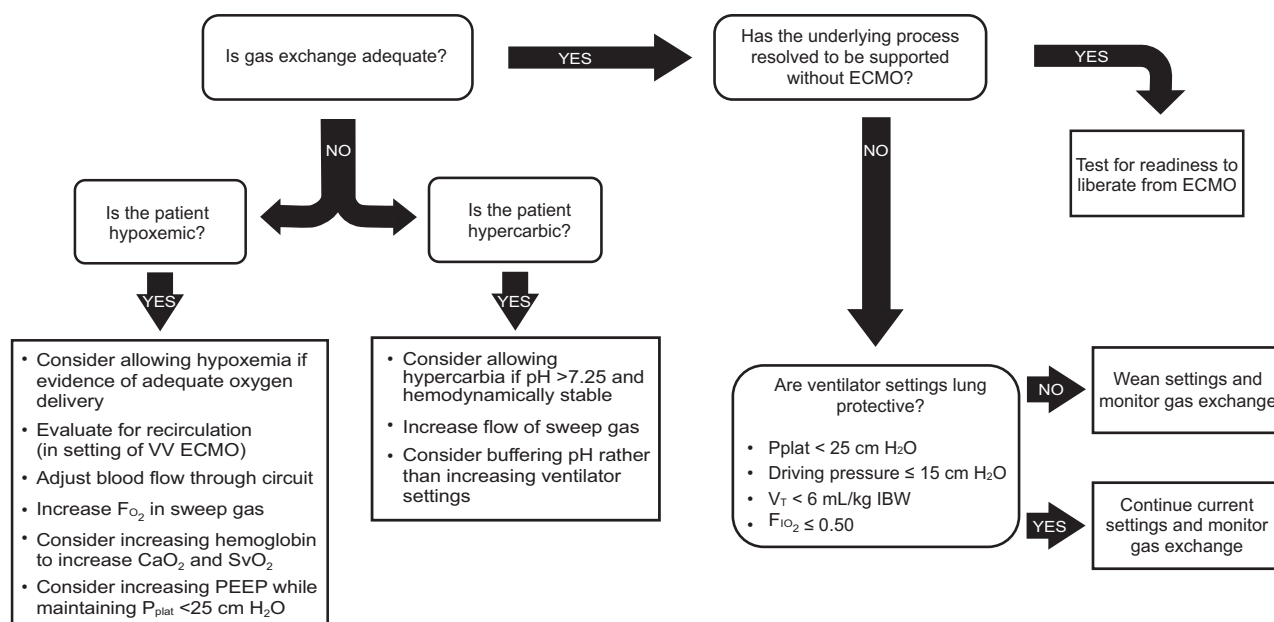


Fig. 2. Algorithm to guide management of gas exchange on extracorporeal membrane oxygenation (ECMO). C_{aO_2} = oxygen carrying capacity; IBW = ideal body weight; S_{vO_2} = mixed venous saturations; VV = veno-venous.

Based on the best available evidence in the literature and anecdotally, we suggest using the algorithm presented in Figure 2 to assist with managing gas exchange on ECMO while maintaining lung-protective ventilation. We also propose the use of pressure control ventilation, preferably as synchronized intermittent mandatory ventilation. Pressure control ventilation is readily available in every ICU regardless of the ventilator type and/or brand available, with most critical care practitioners being very familiar with it. A PEEP of 5–15 cm H₂O (to be titrated to optimize lung expansion if possible without negatively affecting hemodynamics) is reasonable, with most patients ending up at a PEEP of 10–12 cm H₂O, driving pressure/pressure control of 5–10 cm H₂O (targeting $V_T < 4$ mL/kg ideal body weight), with inspiratory plateau pressure < 25 cm H₂O, and intermittent mandatory breaths of 4–15 breaths/min. If synchronized intermittent mandatory ventilation mode is chosen, then a pressure support of 5–15 cm H₂O can be added. As mentioned earlier, patients may tolerate low driving pressures and small V_T while allowing high levels of PEEP without requiring high plateau pressures. In addition to the protective nature of these settings, pressure control ventilation allows practitioners to continue with serial secretion clearance and lung recruitment, and measure dynamic compliance (dynamic compliance = $V_T / [\text{peak inspiratory pressure} - \text{PEEP}]$) regularly to assess for lung improvement.

Non-Traditional Modes of Ventilation

Despite potential theoretical advantages, there have been no consistent data that demonstrate a survival advantage to

any non-traditional mode of ventilation for ARDS. Furthermore, inexpert application of non-traditional modes of ventilation has the potential to worsen lung injury. Use of these modes for patients on ECMO should be done thoughtfully, to address a specific need, and with careful consideration of the knowledge and skill of practitioners across shifts.

Airway pressure release ventilation has the theoretical advantage of using high mean airway pressure to maintain open lungs and assist with oxygenation, which potentially pairs well with a venovenous ECMO circuit that excels at carbon dioxide removal but may not provide full oxygenation needs, if ECMO blood flow is limited by cannula size. The high efficiency of the oxygenator to remove carbon dioxide negates the need for frequent release breaths, which allows more consistent high mean airway pressure. However, the same result can likely be achieved with conventional ventilator settings by using high PEEP and low V_T . Although airway pressure release ventilation has been used successfully for patients on ECMO and was included as an alternative mode of ventilation in the EOLIA study protocol,¹⁸ there are no data that suggest that airway pressure release ventilation during ECMO affects outcomes. High-frequency oscillatory ventilation (HFOV) is another potential mode that uses high mean airway pressure to assist with lung recruitment and oxygenation but also has several downsides for the patient on ECMO. Patients on HFOV commonly require high levels of sedation and paralysis, and secretion removal is often suboptimal. Although HFOV has been used in conjunction with ECMO, it is more typically used as a pre-ECMO rescue therapy, and patients

who require HFOV are transitioned back to conventional ventilation once ECMO is initiated.

High-frequency percussive ventilation, usually in the form of a volumetric diffusive respirator (Percussionaire Corp, Sandpoint, Idaho) has also been used on ECMO, largely for its lung recruitment and secretion mobilization properties. Results of a few small studies suggest that high-frequency percussive ventilation may not only be safely applied to patients on ECMO but that use of high-frequency percussive ventilation may shorten ECMO duration. Michaels et al³⁴ compared 7 adults on ECMO for ARDS secondary to H1N1 influenza who were managed with high-frequency percussive ventilation with other adult survivors of H1N1 on ECMO in the Extracorporeal Life Support Organization registry.³⁴ The researchers found that the 7 patients managed with high-frequency percussive ventilation averaged > 4 fewer days on ECMO than did the Extracorporeal Life Support Organization cohort (8.0 vs 12.3 d; $P = .03$). Another single-center study compared 14 children on ECMO treated with high-frequency percussive ventilation with 22 historical controls on ECMO and conventional ventilation; the researchers observed an increase in 30- and 60-d ECMO-free days, along with a similar but non-significant decrease in the duration of ECMO for survivors.³⁵ Finally, an additional case series of adults described 5 patients who were unable to be weaned from ECMO on high-frequency percussive ventilation; 4 of the 5 patients survived to hospital discharge.³⁶ These studies were all small, non-controlled studies and should be interpreted with caution, yet their findings support high-frequency percussive ventilation as a safe alternative mode of ventilation with potential benefit for selected patients on ECMO.

High-frequency jet ventilation may be used in neonates and infants for similar purposes as high-frequency percussive ventilation, assisting with lung recruitment and mobilization of secretions. This mode of ventilation is most commonly provided through the Bunnell LifePulse ventilator (Bunnell, Salt Lake City, Utah) and is generally limited to infants ≤ 10 kg.³⁷ Similar to HFOV, high-frequency jet ventilation is more commonly used as a rescue therapy for refractory respiratory failure to prevent the need for extracorporeal support, and the patient is typically transitioned back to conventional ventilation once on ECMO. Yet, for certain disease processes, including air-leak syndromes and congenital diaphragmatic hernia, the lung-protective nature of either high-frequency jet ventilation or HFOV may be advantageous to continue while on ECMO.

Adjunct Therapies

Prone positioning has been shown to have multiple physiologic benefits in patients with ARDS, with demonstrated mortality benefit for adults with severe ARDS.^{38,39} It conceptually follows that patients with ARDS and on ECMO might also

have some physiologic benefits from proning, including opening posterior lung segments and improved ventilation/perfusion matching, which improves gas exchange and offloads the right ventricle. Prone positioning has become commonplace for patients with ARDS, including during the COVID-19 pandemic, and this practice has extended to patients with ARDS and on ECMO.²² Despite the potential added complications of proning of patients on ECMO, several case series have demonstrated that proning of patients on ECMO can be done safely with minimal, non-life threatening complications. Only one multi-center study to date has described an outcome difference with subjects in the prone position and on ECMO, reporting a longer ECMO duration but lower mortality (34% vs 50% in the control group; $P = .02$).⁴⁰ Despite a reasonably large sample size of 240 subjects on ECMO, the propensity matching demonstrated a higher Sequential Organ Failure Assessment score for the control patients, and, therefore, the mortality benefit may have been overestimated. A separate meta-analysis supported the safety of proning patients on ECMO but showed no mortality benefit and was associated with a longer duration of ECMO.⁴¹

Inhaled nitric oxide is also known to improve ventilation/perfusion matching in the setting of ARDS but has not been shown to affect long-term outcomes, including mortality in patients with ARDS. In lieu of proven right-ventricular dysfunction and a desire to decrease pulmonary vascular resistance to lower right-ventricular afterload, there is likely no real reason to continue inhaled nitric oxide once the patient is on ECMO. However, it is reasonable to use inhaled nitric oxide to temporarily increase oxygenation in the peri-ECMO cannulation period. It may also be reasonable to consider the use of inhaled nitric oxide to assist with successful decannulation from ECMO but only if the risks of ECMO in that particular patient have overtaken the potential benefit of additional time on ECMO.

Conceptually, glucocorticoids should have a beneficial effect in reducing inflammation that drives biotrauma and contributes to worsening lung compliance; however, the data with regard to steroid use for ARDS treated with ECMO is mixed. A recent multi-center randomized trial and 2 meta-analyses each demonstrated some benefit to steroids for ARDS,⁴²⁻⁴⁴ with increased ventilator-free days and reduced mortality. More recent data may be confounded by dexamethasone as a standard of care for COVID-19 ARDS, yet similar benefits from steroids seem to exist for patients without COVID-19 as well.⁴² Specific to patients with ARDS and on ECMO, administration of steroids for patients with ARDS was associated with improved lung compliance in one study,⁴⁵ yet were also associated with an increase in mortality after successful weaning from ECMO in a review of a multi-center registry.⁴⁶

Neuromuscular blockade may provide benefit early in the course of ARDS, through avoiding ventilator asynchrony

Table 2. Summary of Utility of Ancillary Therapies for Patients on ECMO

Ancillary Therapy	Recommendation	Strength of Evidence
Prone positioning	Safe on ECMO with non-life threatening complications; potential mortality benefit; may increase duration of ECMO	Cohort studies
Inhaled nitric oxide	No indication to continue after initiation of ECMO other than RV dysfunction	Case reports
Glucocorticoids	May improve lung compliance for patients on ECMO but also associated with increased post-ECMO mortality	Observational studies
Neuromuscular blockade	Reasonable to use for safety of the patient, wean as able to help mitigate critical illness myopathy and debility	Not studied in patients on ECMO
Surfactant	Surfactant use in neonatal ECMO may improve survival and reduce ECMO duration; may improve dynamic compliance in pediatric patients with ARDS and on ECMO.	One blinded, randomized control study in neonates, retrospective chart review, case reports
Airway clearance	Bronchoscopy seems to be safe on ECMO; may be some benefit to high-frequency percussive ventilation to reduce ECMO duration in select patients	Case reports and cohort studies

ECMO = extracorporeal membrane oxygenation
RV = right ventricle

and reducing plateau pressure, frequency, and total mechanical work delivered to the lungs. Similarly, neuromuscular blockade will typically be necessary in the peri-cannulation phase of ECMO, with > 40% of patients paralyzed on days 1 and 2 of ECMO.²² Yet neuromuscular blockade has also been associated with critical illness myopathy and polyneuropathy,⁴⁷ which may prolong recovery. Despite earlier data that support a mortality benefit with early neuromuscular blockade for ARDS,⁴⁸ a recent meta-analysis of neuromuscular blockade in subjects with moderate-to-severe ARDS did not demonstrate a mortality benefit.⁴⁹ The use of neuromuscular blockade in ECMO has not been fully studied but may be considered in select patients when necessary to avoid VILI from ventilator dyssynchrony. Neuromuscular blockade should be weaned as soon as possible to allow spontaneous breathing, reduce sedation, and avoid myopathy.

Bronchoscopy may be useful to assist with secretion removal, particularly in cases of patients with plastic bronchitis or significant mucous plugging of airways. Similar to prone positioning, there have been reports that demonstrate bronchoscopy in subjects who are anticoagulated and on ECMO is safe, even in infants and children.⁵⁰⁻⁵² However, to date, there are no data to suggest that routine bronchoscopy provides improved outcomes for patients on ECMO; therefore, clinicians should consider which specific patients on ECMO might benefit from bronchoscopy rather than making it routine practice.

Changes and dysfunction of endogenous surfactant play a role in the pathogenesis of ARDS. Administration of endogenous surfactant may restore surfactant level and function, and

early studies showed promise of endogenous surfactant, including decreased mortality and increased ventilator-free days in pediatric patients with ARDS.^{53,54} However, similar to other adjunct therapies, subsequent studies failed to demonstrate a mortality advantage with endogenous surfactant administration.^{54,55} Endogenous surfactant use for patients on ECMO has been described in neonates and sporadically in pediatric patients. An earlier study showed an improved survival, decrease in neonatal ECMO duration, and improved pulmonary mechanics.⁵⁶ A more-recent pediatric study showed that administration of an endogenous surfactant during pediatric ECMO improved dynamic compliance with no change in survival.⁵⁷ To date, the available data do not support routine use of endogenous surfactant for patients on ECMO. The potential utility of these different ancillary therapies is summarized in Table 2.

Summary

ECMO has the potential to be a lifesaving therapy for refractory hypoxemic respiratory failure, but ultimate survival relies on optimal management of the ventilator to allow lung recovery. Careful attention should be applied to lung-protective strategies, including targeting plateau pressure ≤ 25 cm H₂O, driving pressure ≤ 15 cm H₂O, $V_T < 6$ mL/kg ideal body weight, $F_{IO_2} \leq 0.50$, and ventilator synchrony. ECMO sweep gas should be used liberally to support these lung-protective strategies and to address air hunger. Non-traditional modes of ventilation may have a place in selected patients but should be used thoughtfully

and with the appropriate multidisciplinary expertise for that mode of ventilation.

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